Instructor Performance and Course Quality Evaluation on EduPro

Detailed guide and project requirements for the Instructor Performance and Course Quality Evaluation on EduPro analysis.

Background and Context:

In online education, instructor quality and course design are the strongest drivers of:

- Learner satisfaction
- Course ratings
- Repeat enrollments
- Platform credibility

Even with strong learner demand, poor instructional quality can lead to:

- Low course ratings
- Negative reviews
- Reduced trust in the platform

EduPro needs a data-driven framework to evaluate:

- Instructor effectiveness
- Course quality consistency
- Relationship between teaching expertise and learner satisfaction

Unified Mentor Toronto Government Parks, Forestry & Recreation

Problem Statement

Currently, EduPro lacks clarity on:

- Which instructors consistently deliver high-quality courses?
- Does teaching experience translate into better-rated courses?

- Are some course categories more dependent on instructor quality?
- How evenly is teaching performance distributed across the platform?

Without structured analysis, instructor evaluation remains subjective and fragmented.

Dataset Fields Utilized (High-Dimensional)

Teachers Sheet

- TeacherID
- TeacherName
- Age
- Gender
- Expertise
- YearsOfExperience
- TeacherRating

Courses Sheet

- CourseID
- CourseName
- CourseCategory
- CourseLevel
- CourseRating

Transactions Sheet

- TransactionID
- CourseID
- TeacherID

Key Analytical Questions

This project aims to answer:

- What is the overall distribution of instructor ratings?
- Do instructors with more experience receive higher ratings?
- Is there a relationship between TeacherRating and CourseRating?
- Which expertise areas consistently deliver high-quality courses?
- Are highly rated instructors associated with higher enrollments?

Analytical Methodology (Step-by-Step)

Data Integration

- Join Teachers ↔ Courses ↔ Transactions using TeacherID and CourseID
- Validate mapping between instructors and their courses

Instructor Profile Analysis

- Distribution of instructor age, experience, and expertise
- Rating spread across instructors
- Identification of top-performing and low-performing instructors

Experience vs Performance Analysis

- Correlation between:
  - YearsOfExperience and TeacherRating
  - YearsOfExperience and CourseRating
- Identify diminishing returns or experience thresholds

Course Quality Evaluation

- CourseRating analysis by:
  - CourseCategory
  - CourseLevel
- Gender vs course level comparisons
- Identify categories with consistently high or low ratings

Instructor Impact on Course Success

- Compare course ratings for:
    - High-rated instructors
    - Mid-rated instructors
    - Low-rated instructors
- Enrollment volume comparison by instructor rating tier

Expertise-Based Performance Insights

- Instructor expertise vs course quality
- Identify domains where teaching quality is most critical
- Highlight expertise gaps or training needs

Key Performance Indicators (KPIs)

KPI Name	Description
Average Teacher Rating	Teaching quality benchmark
Average Course Rating	Content effectiveness
Rating Consistency Index	Instructor reliability
Experience Impact Score	Value of teaching tenure
Enrollment Influence Ratio	Instructor-driven demand

Streamlit Web Application Requirements

Core Modules

- Instructor performance leaderboard
- Experience vs rating scatter plots
- Course quality heatmaps
- Expertise-wise performance comparisons

User Capabilities

- Instructor expertise filters
- Course category & level selectors

- Rating range sliders

Deliverables and Submission

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)
- Executive summary for government stakeholders

Conclusion

This project establishes a data-driven instructor and course quality evaluation framework for EduPro. By shifting focus away from learners and toward teaching effectiveness, it enables the platform to identify excellence, address gaps, and continuously improve educational outcomes—making it fundamentally different in purpose.